overwhelmingly points toward its being extremely unlikely. But history surely has taught us that every time our understanding of the universe deepens, we find yet smaller microscopic ingredients constituting a finer level of matter. And so another possibility, should strings fail to be the final theory, is that they are one more layer in the cosmic onion, a layer that becomes visible at the Planck length, although not the final layer. In this case, strings could be made up of yet-smaller structures. String theorists have raised and continue to pursue this possibility. To date there are intriguing hints in theoretical studies that strings may have further substructure, but there is as yet no definitive evidence. Only time and intense research will supply the final word on this question.

Aside from a few speculations in Chapters 12 and 15, for our discussion here we approach strings in the manner proposed in the first answer—that is, we will take strings to be nature's most fundamental ingredient.

Unification through String Theory

Besides its inability to incorporate the gravitational force, the standard model has another shortcoming: There is no explanation for the details of its construction. Why did nature select the particular list of particles and forces outlined in previous chapters and recorded in Tables 1.1 and 1.2? Why do the 19 parameters that describe these ingredients quantitatively have the values that they do? You can't help feeling that their number and detailed properties seem so arbitrary. Is there a deeper understanding lurking behind these seemingly random ingredients, or were the detailed physical properties of the universe "chosen" by happenstance?

The standard model itself cannot possibly offer an explanation since it takes the list of particles and their properties as experimentally measured input. Just as the performance of the stock market cannot be used to determine the value of your portfolio without the input data of your initial investments, the standard model cannot be used to make any predictions without the input data of the fundamental particle properties.6 After experimental particle physicists fastidiously measure these data, theorists can then use the standard model to make testable predictions, such as what should happen when particular particles are slammed together in an accelerator. But the standard model can no